Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad, Natore-6431

Faculty of Electrical and Computer Engineering

Department of Computer Science and Engineering

Lecture Plan

Program:	B.Sc. Engineering in Computer Science and Engineering
Course Code & Title:	MATH-1141 Differential Calculus and Integral Calculus
Semester:	**Summer-2024**
Batch:	**19th**
Level:	1st Year 1st Semester
Credit Hour:	3.00 Hr.
Name & Designation of Teacher:	Md. Zahurul Islam Lecturer, Department of Mathematics
Class Hours:	Monday 12:40 PM-01:35 PM Wednesday 08:15 AM-09:10 AM Thursday 08:15 AM-09:10 AM
Consultation Hours:	Monday: 01:40 PM -2:35 PM Tuesday: 01:30 PM -2:30 PM
E-mail:	zahurul.math.buet@gmail.com
Mobile:	+88 01738333080
Rationale:	To get the basic idea of the rate at which one quantity changes relative to another, understand the concept of the word limit, continuity and differentiability and the relation between the area and antiderivatives and ability to calculate an antiderivative
Pre-requisite (if any):	None

Course Outcomes (CO):

Upon completion the course, the student will be able to:

1.	Understand various elementary function both algebraically and graphically.
2.	Describe the real understanding of the connection between differentiations and integrations and the area and integration and manipulate different integration by various methods.
3.	Apply the differentiation and integration in real world problem.

Course Content:

Module 1: **Limit, Continuity and Differentiability:** Aim and objective of calculus, Function, Domain and Range, Existence of Limit, Continuous Function and its property, Differentiability
Module 2: **Differentiation:** Formula of Differentiation, Derivatives of different kind of functions, Differentiations of explicit and implicit functions, Successive Differentiation, Leibnitz's Theorem and Differentiations using Leibnitz's Theorem

Module 3: **Expansion of Function:** Mean value theorem, Rolle's Theorem, Taylor's Series, Expanding of a Function using Taylor's Series.
Module 4: **Maximum and Minimum:** Increasing and Decreasing of a Function, Turning Point, Point of Inflation, Concave Up and Down, Maximum and Minimum of Different kind of Functions.
Module 5: **Partial Differentiations and Geometric Applications of Differentiations:** Multivariable Function, Partial Differentiations of multivariable functions, Tangent and Normal equations for different kind of function at a point.
Module 6: **Integration:** Formula of integration, Integration by methods of substitution, Integration by parts, Integration of special function, Integration of partial fraction
Module 7: **Definite Integration and Applications of Integration:** Properties of definite integration, Beta and Gamma Function, Successive integration, Evaluate the Arc Length, Surface Area, Area and Volume.

Text Book (s):
1. Calculus (10th edition) – Howard Anton, Irl Bivens and Stephen Dev
2. Differential Calculus – B.C. Das and B.N. Mukherjee
2. Integral Calculus – B.C. Das and B.N. Mukherjee
Reference Book(s):
1. Advanced Engineering Mathematics – H. K. Das

*** Additional reading material:** PowerPoint slides, Docx & PDF provided by the Teacher.

Course Assessment Methods & their Weights:

Assessment Method	**(%)**
Class participation/Presentation skill	05
Class performance	05
Homework/Assignment	05
Class Tests	15
Semester Final Exam (Maximum 03 Hours)	70
Total:	**100**

Minimum Attendance: 80% class attendance is mandatory for a student in order to appear at the final examination.

Student's responsibilities:

- **a.** Students must come to the class prepared for the course material covered in the previous class.
- **b.** They must submit their assignments on time.
- **c.** No late or partial assignments will be acceptable.
- **d.** There will be no make-up Class Test.

Grading Policy:

Numerical Score	Letter Grade		Grade points
80% and above	A+	A (Plus)	4.00
75% to > 80%	A	A (Regular)	3.75
70% to > 75%	A-	A (Minus)	3.50
65% to > 70%	B+	B (Plus)	3.25
60% to > 65%	B	B (Regular)	3.00
55% to > 60%	B-	B (Minus)	2.75
50% to > 55%	C+	C (Plus)	2.50
45% to > 50%	C	C (Regular)	2.25
40% to > 45%	D	D (Pass)	2.00
below 40%	F	Fail	0.00
Incomplete	I	-	-
Withdrawal	W	-	-
Project / Thesis Continuation	X	-	-

*I-Incomplete; *W-Withdrawn; *X-Projects/Thesis Continuation; *E-Expel

*The subject in which the student gets F grade shall not be counted towards credit hours requirements and for the calculation of Grade Point Average (GPA). Students failing in a course/ courses due to being expelled, dis-collegiate, absent will be treated as F according to Grading system.

Generic Skills: Graduate Attributes/Program Outcomes (POs)

Program Outcomes (POs) represent the knowledge, skills and attitudes the students should have at the end of a four years engineering program. CSE program of BAUET has 12 Program Outcomes. They are briefly described in the following table.

Sl.No	PO	Category	Description
1	**PO 1**	**Engineering Knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2	**PO 2**	**Problem Analysis**	Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3	**PO 3**	**Design/Development of Solutions**	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, social and environmental concerns.
4	**PO 4**	**Investigation**	Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
5	**PO 5**	**Modern Tool Usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6	**PO 6**	**The Engineer and Society**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7	**PO 7**	**Environment and Sustainability**	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of need for sustainable development.
8	**PO 8**	**Ethics**	Apply ethical principles and commit to professional Ethics and responsibilities and norms of the engineering practice.
9	**PO 9**	**Individual and Team Work**	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10	**PO 10**	**Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**PO 11**	**Project Management and Finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	**PO 12**	**Life Long Learning**	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

POs	Graduate Attributes (Keywords)	WA-WK's	WP/EA
1	Engineering Knowledge	WK1-WK4	WP
2	Problem Analysis	WK1-WK4	WP
3	Design/Development of Solutions	WK5	WP
4	Investigation	WK8	WP
5	Modern Tool Usage	WK6	WP
6	The Engineer and Society	WK7	WP
7	Environment and Sustainability	WK7	WP
8	Ethics	WK7	--
9	Individual and Team work	--	--
10	Communication		EA
11	Individual and Team Work	---	--
12	Project Management and Finance	--	--
Assessments provide adequate feedback to the program to identify strengths and weaknesses for CQI			

Complex Engineering Problems (Need High Taxonomy Level)

Complex Engineering Problems have characteristic WP1 and some or all of WP2 to WP7, EP1 and EP2, that can be resolved with in-depth forefront knowledge

WP1	Depth of Knowledge required	Resolved with **forefront in-depth** engineering knowledge (WK3, WK4, WK5, WK6 or WK8) which allows a fundamentals-based, first principles analytical approach
WP2	Range of conflicting requirements	Involve **wide-ranging or conflicting** technical, engineering and other issues.
WP3	Depth of analysis required	Have **no obvious solution** and require abstract thinking, originality in analysis to formulate suitable models.
WP4	Familiarity of issues	Involve **infrequently encountered** issues
WP5	Extent of applicable codes	**Beyond codes** of practice
WP6	Extent of stakeholder involvement and level of conflicting requirements	Involve **diverse groups of stakeholders** with widely **varying needs**.
WP7	Interdependence	Are **high level problems** including **many component** parts or sub-problems.
EP1	Consequences	Have **significant consequences** in a range of contexts.
EP2	Judgement	Require judgement in **decision** making

Complex Engineering Activities (Project based)

Activities	Preamble	**Complex activities** means (engineering) activities or projects that have **some or all** of the following characteristics listed below
EA1	Range of resources	**Diverse resources** (people, money, equipment, materials, information and technologies).
EA2	Level of interaction	Require resolution of significant problems arising from interactions between **wide ranging** or **conflicting** technical, engineering or other issues.
EA3	Innovation	Involve creative use of engineering principles and **research-based** knowledge in **novel** ways.
EA4	Consequences to society and the environment	Have **significant consequences** in a **range of contexts**, characterised by **difficulty** of prediction and mitigation.
EA5	Familiarity	Can extend **beyond previous** experiences by applying **principles-based** approaches.

Bloom's Taxonomy Domain:

Cognitive Domain		Psychomotor Domain		Affective Domain	
C1	Remembering	**P1**	Perception	**A1**	Receive
C2	Understanding	**P2**	Set	**A2**	Respond
C3	Applying	**P3**	Guided Response	**A3**	Value
C4	Analyzing	**P4**	Mechanism	**A4**	Organize
C5	Evaluating	**P5**	Complex Overt Response	**A5**	Internalize
C6	Creating/Designing	**P6**	Adaption		
		P7	Origination		

Detailed Course Plan and Mapping (According to BAETE Accreditation Manual- II)

i. Mapping CO-PO, Bloom's Level, Knowledge Profile, Complex Engineering Problem (WP) and Complex Engineering Activities (EA)

Course No & Title: MATH-1141 (Differential Calculus and Integral Calculus)

COs	CO Statements	PO	KP	Bloom's Level	Delivery methods and activities	Assessment Tools
CO1	Understand various elementary function both algebraically and graphically.	1	1	C1, C2	Lectures, Text Book, PPT, PDF White Board Writing	Class Participation/ Observation Homework/ Assignment, Class Test, Semester Final Examination (Written)
CO2	Describe the real understanding of the connection between differentiations and integrations and the area and integration and manipulate different integration by various methods.	2	2	C2, C3		
CO3	Apply the differentiation and integration in real world problem.	2	2, 3	C2, C3		

Legends: CO-Course Outcomes PO-Program Outcomes, KP-Knowledge Profile
Bloom's Level: Cognitive (C1-C6), Psychomotor(P1-P7), Affective(A1-A5)

Course Outcomes (CO)	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1	H											
CO2		M										
CO3		M										

Legends: High(H)=3, Medium(M)=2 and Low(L)=1

a. Assessment of CO-PO, Bloom's Level, Knowledge Profile, Complex Engineering Problem (WP) and Complex Engineering Activities (EA)

Teaching-learning and Assessment Strategy:
Lectures, Class Participation, Homework/Assignments, Class Tests and Semester Final Exam (Written)

Assessment Methods	CO	PO	Bloom's Level	KP	WP & EA	Remarks
Class Test	1 3	1 2	C1, C2 C3	1-2	--	Question **1** from Section **A** will satisfy specified CO-POs, BLs, KPs and WPs
Homework/Assignment	1,3	1,2	C3	1-2		
Semester Final Exam (Written)	1 2 3	1 2	C1, C2, C3	1-2		

Teaching Plan:

Module-1: Limit, Continuity and Differentiability:

Learning Outcomes:
After completion of this module, student should be able to:

1. Ability to demonstrate any function and identify a function at a point, whether it is continuous or not.
2. Ability to identify a function at a point, whether it is differentiable or not.

Week & Class No.	Topic to be covered	CO
W-1,C-1	Introduction to function and illustrations of different kind of functions. Finding the domain and range of different kind of functions.	CO1
W-1,C-2	Condition for existing limit of a function. Whether exists limit of a function and evaluate the limiting value of different kind of function at a point.	CO1
W-1,C-3	Formulas for limit and exercises of limiting value problems.	CO1
W-2,C-4	Conditions for continuity of a function at a point. Identify whether a function is continuous or not, at a point.	CO1
W-2,C-5	Exercises of continuity problems.	CO1
W-2,C-6	Conditions for differentiability of a function at a point. Identify whether a function is differentiable or not, at a point.	CO1
W-3,C-7	Exercises of differentiability problems.	CO1
W-3,C-8	**Quiz Test**	

Application Areas:

1.	Limit, continuity, and differentiability are used to attribute different functions that arises from real life problems.

Review Questions:		CO
1.	What do you mean by Limit?	CO1
2.	State the conditions of existing limit, continuity and differentiability.	CO1
3.	Does limit exist of the function $f(x) = \|x\|$ $at\ the\ point\ x = 0$ and evaluate $\lim_{x\to 0} \frac{\sin x^5}{5x}$	CO1
4.	Show that $f(x) = \|x\|$ is continuous but not differentiable at $x = 0$.	CO1

Module-2: Differentiations:

Learning Outcomes:

After completion of this module, student should be able to:

1. Differentiate different kind of function.
2. Apply differentiations in real life problem like finding velocity, accelerations etc.

Class No	Topic to be covered	CO
W-3,C-9	Formulas of differentiations and proof of some formulas	CO2
W-4,C-10	Evaluate the differential coefficient of different kinds of functions	CO2
W-4,C-11	Differentiations of various composite functions	CO2
W-4,C-12	Formula and Exercises of differentiations of the product and division of two or more functions	CO2
W-5,C-13	Technic to evaluate differential coefficients using logarithm and some exercises	CO2
W-5,C-14	Finding the differentiations of implicit functions.	CO2
W-5,C-15	**CT-01**	CO2
W-6,C-16	Technic of successive differentiation and calculation of successive differentiations of algebraic and trigonometric	CO2
W-6,C-17	Finding the successive differentiations of logarithmic, exponential functions and function with algebraic fraction.	CO2
W-6,C18	Statement of Leibnitz's theorem and finding the n-th derivatives of the product of two functions using Leibnitz's theorem	CO2

Application Areas:

1.	To find rate of change of a quantity
2.	To find the approximation value

Review Questions:		**CO**
1.	Prove that $\frac{de^x}{dx} = e^x$	CO2
2.	Find the differential coefficients of $\sin ax$, $\tan bx^2$, $ax^2 + bx + c$, lnx^3	CO2
3.	Differentiate the following functions: $\frac{1+\csc(x^2)}{1-\cot(x^2)}$, $\cos^2(3\sqrt{x})$, $\sin(\sqrt{1+\cos x})$, $a^{\sin x}, e^{\tan x}, e^x \tan x$	CO2
4.	Find $\frac{dy}{dx}$ from the following implicit functions: $(\sin y)^x = (\cos x)^y, \sqrt[x]{\sin^{-1} y} = \sqrt[y]{\cos^{-1} x}$	CO2

5	If $y = \cos(m \sin^{-1} x)$ then show that (i) $(1-x^2)y_2 - xy_1 - m^2 y = 0$ (ii) $(1-x^2)y_{n+2} - (3n+1)xy_{n+1} - (n^2 - m^2)y_n = 0$. Also find y_n when $x = 0$.	CO3

Module-3: Expansion of Function:

Learning Outcomes:

After completion of this module, student should be able to:

1. Expand different kinds of special functions in polynomial series.
2. Approximate function at a point using Taylor's series..

Class No	Topic to be covered	CO
W-7,C-19	Roll's Theorem, Mean Value Theorem and Generalized Mean Value Theorem	CO2
W-7,C-20	Expansion and approximate of transcendental function using Taylor series	CO2
W-7,C-21	**CT-02**	CO2

Application Areas:

1.	Expanding the transcendental function to polynomial series.
2.	Approximation of a function at a point using Taylor series

Review Questions:		CO
1.	State Rolle's Theorem, Mean Value Theorem and Generalized Mean Value Theorem.	CO2
2.	Approximate e^x at $x = 0$	CO2

Module-4: Maximum and Minimum:

Learning Outcomes:

After completion of this module, student should be able to:

1. Identify whether a function is increasing or decreasing at a point.
2. Find the turning point and point of inflation of different kind of function
3. Evaluate both local and global maxima and minima of different kind of function.

Class No	Topic to be covered	CO
W-8,C-22	Identify a function is increasing or decreasing at a point. Also finding the turning point.	CO2
W-8,C-23	Evaluate the local and global maximum and minimum of a function and some exercises.	CO2
W-8,C-24	Finding the point of inflation and identify the range whether the function is concave up or down.	CO2

Application Areas:

1.	Finding the nature of different kind of function like turning point, maxima, and minima.
2.	Applicable in physical problems like maximum area of a crop land with the fixed length of fence.

Review Questions:		CO
1.	Find the domain of a function $f(x) = x^2 + 2$ whether it is increasing or decreasing.	CO2
2.	Evaluate the turning point and identify whether it has maximum or minimum value of the function $f(x) = x^2 + 2$.	CO2
3.	Identify the domain whether a function $f(x) = x^3 - 2x + 3$ is concave up or down. Also find the point of inflations.	CO2

Module-5: Partial Differentiations and Geometric Applications of Differentiations:

Learning Outcomes:

After completion of this module, student should be able to:

1. Evaluate the rate of change with respect to one variable of a multivariable function.
2. Ability to connect differentiation with real life.

Week & Class No.	Topic to be covered	CO
W-9,C-25	Introduce to multivariable function and partial derivatives and it's properties.	CO2
W-9,C-26	Evaluate partial derivative of different kind of multivariable functions.	CO2
W-9,C-27	Geometric Applications of ordinary and partial derivatives.	CO3
W-10,C- 28	**Assignment**	CO3

Application Areas:

1.	Finding the tangent planes and Linear Approximations.
2.	Evaluating the gradient vector and normal lines.

Review Questions:		CO
1.	What do you mean by multivariable function?	CO2
2.	Evaluate the partial derivative with respect to the variables of the following multivariable function: $f(x,y,z) = 3xy^2z + 5x^2yz - 3z^3$	CO2

Module-6: Integration:

Learning Outcomes:

After completion of this module, student should be able to:

1. Evaluate the indefinite integrals.
2. Relate between the derivatives and anti-derivatives.

Class No	Topic to be covered	CO
W-10,C-29	Introduce to integration and it's physical meaning and formulas of integrations	CO1
W-10,C-30	Integration by methods of substitution and exercises	CO2
W-11,C-31	Integration by parts and it's exercises	CO2
W-11,C-32	Integration of special function	CO2
W-11,C-33	Integration of partial fraction and exercises	CO2
W-12,C-34	Revision Class	CO2
W-12,C-35	**CT-03**	

Application Areas:

1.	Integrations are used in various fields such as engineering, where engineers use integrals to find the shape of building.
2.	In physics, there are many applications of integration like the center of gravity etc.

Review Questions:		**CO**
1.	White down the physical interpretations of integrations.	CO2
2.	Evaluate the integrations: (i) $\int [x^{-1/2} - 3x^{7/5} + \frac{1}{9}]\,dx$ (ii) $\int x^{1/3}(2-x)^2\,dx$ (iii) $\int \sin^2 x \cos x\,dx$.	CO2

Module-7: Definite Integration and Applications of Integration:

Learning Outcomes:

After completion of this module, student should be able to:

1. Evaluate the integral value within a fixed domain
2. Apply integration in real life problems like finding length, area, volume etc.

Class No	Topic to be covered	CO
W-12, C-36	Introduction to definite integral and it's properties	CO1
W-13, C-37	Exercises of definite integrals	CO1
W-13, C-38	Introduce to the Beta and Gamma Function and Successive integrations	CO2
W-13, C-39	Evaluate the Arc Length and exercises	CO3
W-14, C-40	Evaluate the Surface Area, Area and Volume.	CO3
W-14, C-41	Review Class-I	CO1- CO2
W-14, C-42	Review Class-II	CO2- CO3

Application Areas:

1.	Geometric applications of definite integrals are evaluating the arc length, surface area and volume etc.

2.	Physical applications of definite integrals are evaluating the mass and work, to growth and decay models.

Assignment		**CO**
1.	A projectile is fired vertically upward with an initial velocity of 49 m/s from a tower 150 m high. (a) How long will it take for the projectile to reach its maximum height? (b) What is the maximum height? (c) How long will it take for the projectile to pass its starting point on the way down? (d) What is the velocity when it passes the starting point on the way down? (e) How long will it take for the projectile to hit the ground? (f) What will be its speed at impact?	CO3
Review Questions:		
1.	State the Fundamental Theorem of Calculus.	CO1
2.	Evaluate the definite integrals: (i) $\int_{-1/2}^{1/2} \frac{1}{\sqrt{1-x^2}}\,dx$ (ii) $\int_0^{\pi/3} (2x - \sec x \tan x)\,dx$ (iii) $\int_0^{\pi/8} \sin^5 2x \cos 2x\,dx$	CO2
3.	In each part, use the given information to find the position, velocity, speed, and acceleration at time t = 1 (a) $v = \cos\frac{\pi t}{3}; s = 0\ when\ t = \frac{3}{2}$ (b) $a = 4e^{2t-2}; s = \frac{1}{e^2}\ and\ v = \left(\frac{2}{e^2}\right) - 3\ when\ t = 0$	CO3

Prepared by:

Md. Zahurul Islam
Lecturer,
Dept. of Mathematics,
BAUET

Approved by:

Mohammed Golam Sarwar Bhuyan
Professor and Head
Dept. of CSE, BAUET